

NAONAO ZHONG

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🎓: Undergraduate in Data Science 📍: Shanghai, China 🔍: Computational Neuroscience

Education

Shenzhen Middle School, <i>High School</i>	2021.9 - 2024.6
Shanghai Medical College, Fudan University, <i>Undergraduate</i> , Clinical Medicine	2024.9 - 2029.6
School of Data Science, Fudan University, <i>Undergraduate</i> , Data Science	2026.9 - 2029.6
• GPA: 3.97 Rank: 4/212	
• English Proficiency: CET4: 646 CET6: 624	

Research Interests

- **Computational Neuroscience:** hierarchical processing and transmission of information in the brain
- **Neural Computation:** identifying the representational level most suitable for computational modeling
- **AI for Neuroscience:** leveraging large language models, reinforcement learning, and other emerging methods to model and understand brain function

Research Experience

Mechanisms and Applications of GABA Release Patterns 2025.9 - present

Advisor: Prof. Yousheng Shu, Institute for Translational Brain Research, Fudan University

- **Scientific Question:** DBS therapy for parkinsonian motor deficits correlates with stimulation frequency and asynchronous release (AR) at STN GABAergic synapses; how DBS parameters modulate AR and improve motor function remains unclear.
- **Behavioral:** Compared high-frequency (130 Hz), low-frequency (20 Hz), and alternating trains, revealing differential effects on motor performance.
- **Electrophysiology:** Applied optogenetic/electrical stimulation in acute STN slices and performed whole-cell patch-clamp recordings of IPSCs. Quantified stimulation-dependent changes in the AR/SR balance, suggesting a potential role of asynchronous inhibition in DBS efficacy.
- **Computational:** Modeled synaptic release to investigate calcium-dependent AR mechanisms and simulate their impact on network dynamics.

Neural Signal Separation via Template Deconvolution 2026.2

- **Objective:** Automatic extraction and sorting of single-neuron spikes from large-scale neural recordings.
- **Data:** High-density silicon probes (e.g., Neuropixels) and two-photon calcium imaging.
- **Methodology:** Convolutional generative model with low-rank spatiotemporal templates and sparse firing trains. Alternating optimization via matched-filter detection, spike-triggered averaging, and truncated SVD under a Bayesian framework.
- **Implementation:** Implemented differentiable convolution operators in PyTorch, including cross-correlation and matrix factorization modules. Validated the pipeline on SpikeInterface synthetic data and real electrophysiology recordings, with design references from Kilosort.

Neural-to-Phoneme Decoding for Intracortical Speech BCI 2026.5

- **Objective:** Decode natural language from intracortical neural signals.
- **Data:** Intracortical recordings from the *BrainToText Benchmark*, where a participant attempted to speak or mouth prompted sentences while neural activity was acquired via a 256-channel microelectrode array.
- **Methodology:** A two-stage framework: neural signals are first decoded into phoneme sequences and then translated to text. This work focuses on the neural-to-phoneme decoding stage.
- **Implementation:** Extracted and normalized multi-channel neural features from intracortical recordings for sequence-to-sequence decoding, with block-wise normalization, feature selection, and HDF5 serialization for efficient storage. Built a bidirectional GRU with CTC loss in PyTorch, supporting variable-length sequence packing and dynamic batching.

Honors and Awards

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| • First Prize in National High School Mathematics League | 2023.11 |
| • Undergraduate Second Class Scholarship | 2025.12 |